

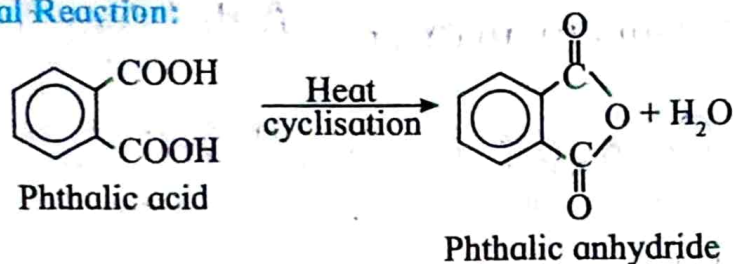
Activity No.1

Date: / /

Aim: To prepare phthalic anhydride.

Theory: Phthalic acid is a dicarboxylic acid which on heating loses a water molecule and undergo cyclisation reaction to form phthalic anhydride.

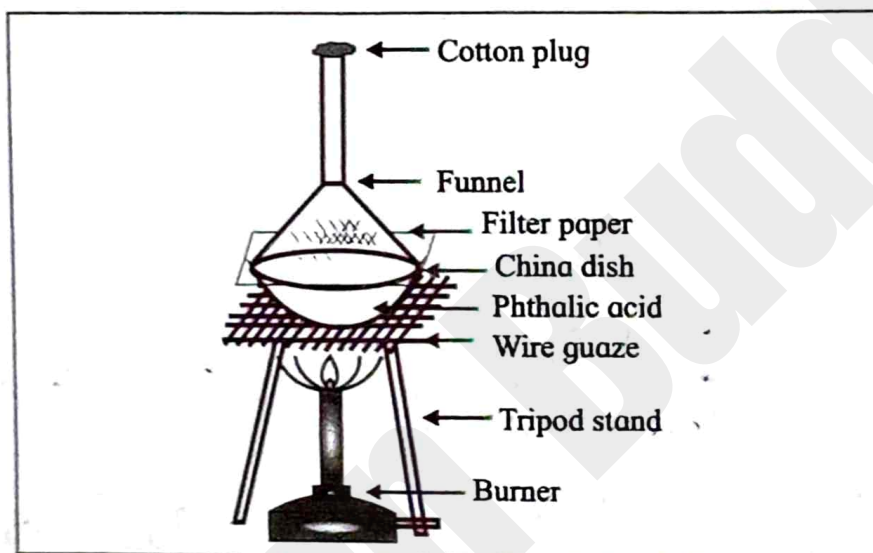
Chemical Reaction:



Apparatus: Evaporating dish, funnel, cotton, tripod stand, wire gauze, pair of tongs, filter paper, etc.

Chemicals: Phthalic acid.

Diagram:



Procedure:

1. Take 2 g of phthalic acid in a dry evaporating dish.
2. Cover the dish with filter paper having number of small holes in the centre.
3. Keep an inverted funnel on the filter paper and close nozzle of the funnel with cotton plug.
Place evaporating dish on the tripod stand with wire gauze as shown in above diagram.
4. Now heat the china dish slowly and carefully on low flame.
5. After sometime the acid sublimes and anhydride collects on the inner side of the funnel.
6. Stop heating and remove the funnel carefully with pair of tongs and cool.
7. Collect the crystals of phthalic anhydride on the paper and weigh it.

Result:

1. Colour of the crystals = white
2. Shape of the crystals = needle
3. Yield of phthalic anhydride = 1.78 g

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

1. Phthalic acid is dicarboxylic acid
 - a. monocarboxylic acid
 - ☒ b. dicarboxylic acid
 - c. tricarboxylic acid
 - d. polycarboxylic acid
2. Phthalic anhydride is obtained by the process known as ____
 - a. decomposition
 - ☒ b. dehydration
 - c. sublimation
 - d. distillation
3. Phthalic acid undergo ____ during preparation of phthalic anhydride
 - ☒ a. dehydration
 - b. decarboxylation
 - c. dehalogenation
 - d. dehydrohalogenation
4. Shape of crystals of phthalic anhydride is ____
 - ☒ a. needle
 - b. hexagonal
 - c. triclinic
 - d. monoclinic
5. Formation of phthalic anhydride takes place by ____ process/reaction.
 - a. isolation
 - ☒ b. cyclisation
 - c. decomposition
 - d. diazotization

Short answer questions

1. Why cotton plug is used in the process of sublimation?

Ans. Cotton plug is used to stop the sublimated compound inside the funnel.

2. Write chemical equation for preparation of acetic anhydride from acetic acid.

Ans.
$$\begin{array}{c} \text{CH}_3\text{-COOH} \xrightarrow{\Delta} \text{CH}_3\text{-CO} \\ \text{acetic acid} \quad \quad \quad \text{Acetic anhydride} \end{array}$$

3. What is the colour and shape of phthalic anhydride crystals?

Ans. Colour - white
Shape - needle shape.

4. Why slow heating is required to the contents during process of sublimation?

Ans. During sublimation, at low temperature solid substance gets converted into gaseous state.

5. Why filter paper is used with small holes in the centre during preparation of phthalic anhydride?

Ans. Phthalic acid when heated sublimes. Vapour of phthalic anhydride passes through holes of the filter paper and gets condensed in the funnel.

Remark and sign of teacher: